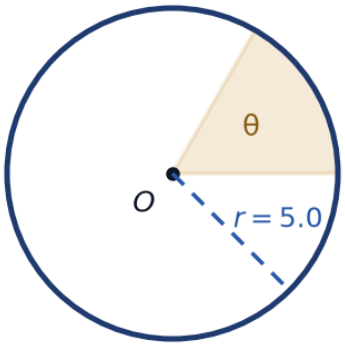
Permutations & Combinations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 16.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

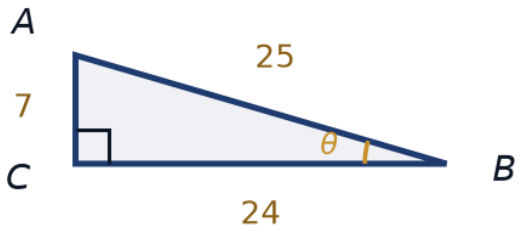
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle B and the corresponding values of side c.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Three mobile antennas are located at coordinates $A(0, 0)$, $B(6, 0)$, and $C(2, 4)$. (a) Find the equation of the perpendicular bisector of segment AB. (b) Find the equation of the perpendicular bisector of segment AC. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

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4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

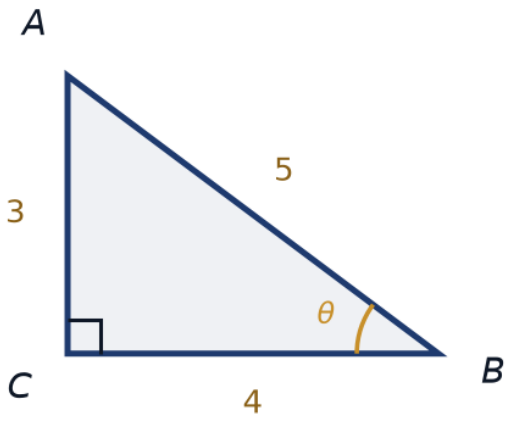
Six wines were evaluated by two independent sommeliers, X and Y, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient r_s. (c) State what this coefficient indicates about the judges' agreement.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A curve has equation $f(x) = \frac{2x^2}{x^2 + k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of k. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of k.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

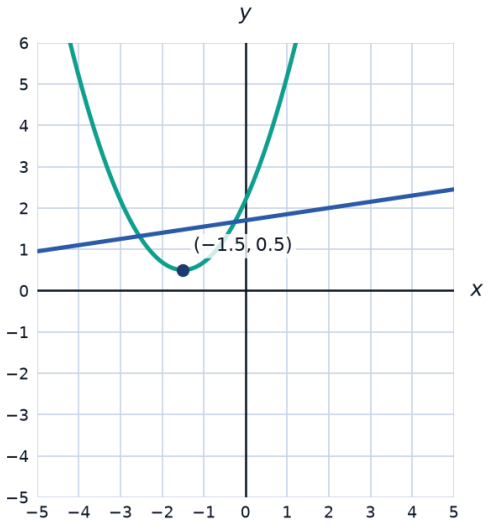
A closed cylindrical metal container with circular base of radius r cm and height h cm has a fixed volume of 250π cm^3. (a) Express h in terms of r. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius r that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A continuous random variable X is normally distributed with mean μ and standard deviation σ. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of μ and σ correct to 3 significant figures.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the function $f(x) = x^2 e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of f. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Determine whether the lines $\mathbf{r}_1 = (1; 2; -1) + \lambda(2; 1; 3)$ and $\mathbf{r}_2 = (0; 5; 4) + \mu(-1; 2; 1)$ intersect or are skew.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

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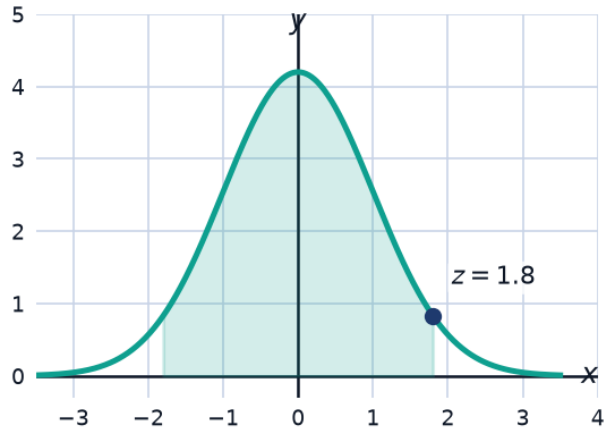
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 12.01 Forces & Equilibrium

Practice AI.1: Use the trapezoidal rule with 4 subintervals to estimate $\int_1^3 \sqrt{1+x^3}\,dx$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 13.01 Newton's Laws

Consider the points $A(1,2,3)$, $B(2,4,1)$, and $C(-1,0,5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 14.01 Energy, Work & Power

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Representation of Data

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $x \neq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Practice 1.1: In an arithmetic sequence, the 4th term is 19 and the sum of the first 10 terms is 290. Find the first term u_1, the common difference d, and the sum of terms from u_{11} to u_{25} inclusive.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 18.01 Binomial Distribution

Use the substitution $u = 2x+1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many 4-digit numbers can be formed using the digits $1, 2, 3, 4, 5, 6$ without repetition?

a. 360

b. 720

c. 120

d. 240

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 7 runners finish 1st, 2nd, and 3rd in a race, assuming no ties?

a. 210

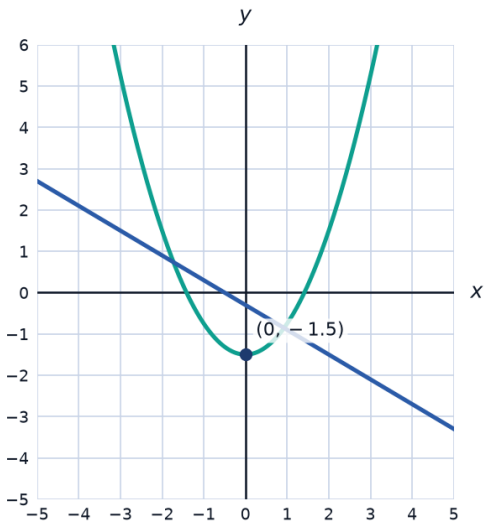
b. 35

c. 5040

d. 120

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many even 3-digit numbers can be formed using digits $1, 2, 3, 4, 5$ without repetition?



a. 24

b. 12

c. 36

d. 60

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many distinct arrangements of all the letters in the word MATHEMATICS are possible?

a. $4,989,600$

b. $39,916,800$

c. $1,247,400$

d. $2,494,800$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 5 boys and 4 girls stand in a line if all 4 girls must stand together?

a. $17,280$

b. $2,880$

c. $43,200$

d. $5,760$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 6 people sit around a circular table, where rotations are considered equivalent?

a. 120

b. 720

c. 24

d. 360

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A committee of 5 is to be chosen from a group of 8 men and 6 women. How many committees are possible if it must contain exactly 3 men and 2 women?

a. 840

b. 420

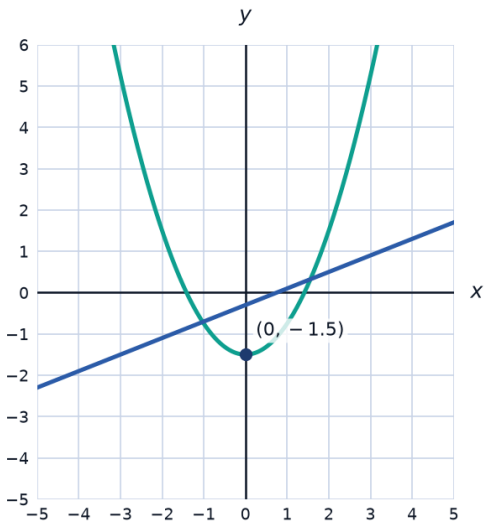
c. 1260

d. 2002

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

From a group of 10 students, a team of 4 is selected. If the team must include the student council president, how many distinct teams are possible?



a. 84

b. 210

c. 126

d. 72

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A team of 4 is selected from 7 boys and 5 girls. How many teams have at least one girl?

a. 460

b. 495

c. 420

d. 35

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How many 5-digit odd numbers greater than 40,000 can be formed using digits $2, 3, 4, 5, 7$ without repetition?

a. 28

b. 36

c. 48

d. 24

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A delegation of 4 is chosen from 5 teachers and 6 students. In how many ways can this be done if there must be more teachers than students?

a. 75

b. 65

c. 85

d. 50

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Four cards are drawn from a standard 52-card deck. What is the probability that all four cards are of the same suit?

a. $\frac{44}{4165} \approx 0.01056$

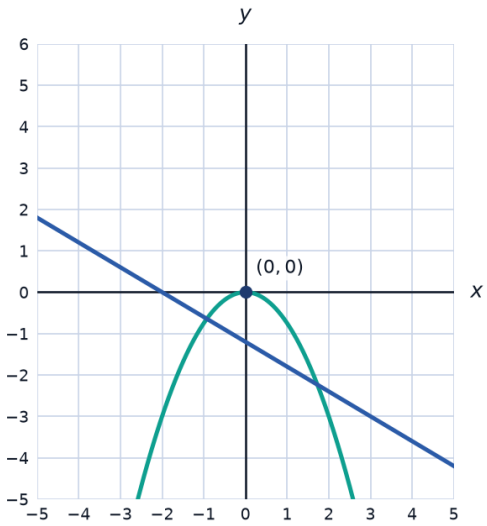
b. 0.0025

c. 0.0450

d. 0.0210

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In how many ways can 11 different people be arranged in a row if two particular people must NOT sit next to each other?



a. 32806546

b. 33364823

c. 34615086

d. 32659200

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In how many ways can 8 different people be arranged in a row if two particular people must sit together?

a. 10080

b. 8546

c. 9551

d. 9288

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In how many ways can 6 different runners be lined up at the start of a race if two particular runners must stand next to each other?

a. 273

b. 240

c. 220

d. 241

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many ways can 5 of 7 different people be selected and arranged in a row?

a. 2891

b. 2282

c. 2520

d. 2348

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A discrete random variable X has distribution $P(X=1)=2/5, P(X=2)=3/10, P(X=3)=3/10$. State $P(X=2)$.

a. 3/4

b. 8/9

c. 3/10

d. 11/15

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 15.01 Representation of Data

Convert 270° to radians, giving your answer in terms of π.

a. $\frac{\pi}{2}$

b. π

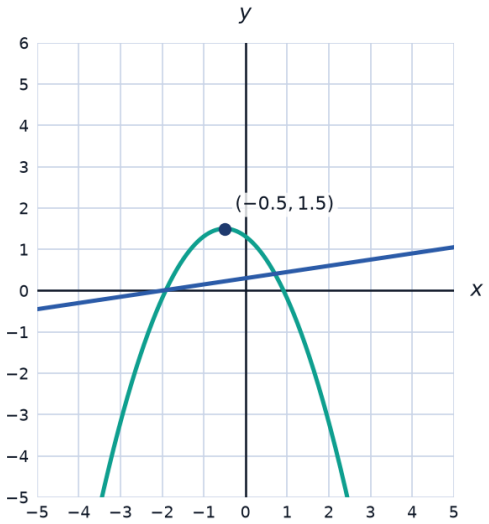
c. $\frac{3\pi}{2}$

d. $\frac{\pi}{6}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 17.01 Discrete Random Variables

Given $X \sim B(6, 3/10)$, find $P(X \geq 1)$, correct to 4 decimal places.



a. 0.8824

b. 0.8324

c. 0.5798

d. 0.1176

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 18.01 Binomial Distribution

A random variable $X \sim B(51, 0.4)$ is approximated by a normal distribution. State the variance of the approximating normal distribution.

a. 12.24

b. 15.24

c. 9.24

d. 19.24

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 19.01 Normal Distribution

Convert $\frac{3\pi}{2}$ radians to degrees.

a. 30°

b. 180°

c. 90°

d. 270°

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the exact value of $1\sin(270^\circ)$.

a. -1

b. -3

c. $-\frac{1}{2}$

d. 1

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Graphs & Transformations

Find the gradient of the tangent to $y = 3x^2 - 6x$ at $x = 3$.

a. 13

b. -12

c. 11

d. 12

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Given $X \sim N(73, 4^2)$, find the standardised value z corresponding to $x = 65$.

a. 2

b. -1

c. -3

d. -2

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Trigonometry

Find the x-coordinate of the stationary point of $y = 3(x + 1)^2 - 2$.

a. 1

b. 0

c. -1

d. -2

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